Does sentiment value impact stock prices?

344 – Data Science and Machine Learning

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Problem description

Application domain

- Financial data analysis
- Stock market behavior
- Taking Sentiment data from Headline news.

Problem

Whether or not sentiment data has a positive/negative impact on the stocks prices for the following day?

Quantatizing sentiments.

Importance

- Understanding the impact of public perception on financial markets
- Imformed trading decisions/ Trading strategies
- Understanding market behaviors

Data Set overview

S&P 500 with Financial News Headlines (2008–2024)

Source: Kaggle

- Each record represent: one financial news headline paired with the daily closing value of the S&P 500 index for that same date.

Number of Samples and Features

- Samples (Rows): 19,000+ news articles
- Features (Columns): 3 variables

Types of feature

- **Text Features:** Headline
- **Numerical Features:** Close price
- **Time-Series Feature:** Date

Data Set overview

yahoo finance api

Source: yfinance

- Each record represent: Each trading day of S&P 500

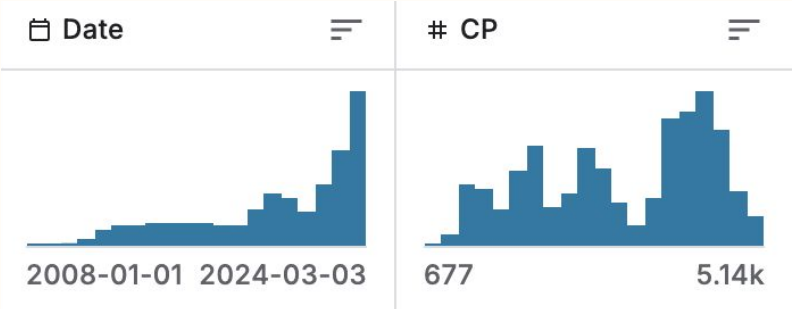
Number of Samples and Features

- Samples (Rows): 4,059 days
- Features (Columns): 7

Types of feature

- **Numerical Features:**
  - Open, High, Low, Close, Adjusted Close, Volume
- **Time-Series Feature:**
  - date and year

Exploratory Data Analysis (EDA)



	date	sent_same_day	neg_same_day	neu_same_day	pos_same_day
0	2008-01-22	0.0000	0.000000	1.000000	0.000
1	2008-01-29	0.0000	0.000000	1.000000	0.000
2	2008-01-30	0.5994	0.000000	0.562000	0.438
3	2008-02-01	0.0772	0.000000	0.755000	0.245
4	2008-02-05	-0.1405	0.119667	0.880333	0.000

Key observation, Quality issues, Key insights

- Most sentiments are neutral
- Multiple headlines occur on the same trading day
- Closing price is really unevenly distributed.

Data Preprocessing and Feature Engineering

Data Cleaning

The two data sets have different format and features.

- Consistent date formats
- Match headlines to trading days
- Removed missing rows

Feature Engineering (technical features)

Not using raw prices. Creating indicators that summarize short-term market behavior.

- 1 day return
- 5 day return
- 10 day return
- 10 day moving average deviation
- 5 day volatility
- 1 day volume change

Sentiment Features (Vader)

Converting headlines into numbers

- Positive score
- negative score
- Neutral score

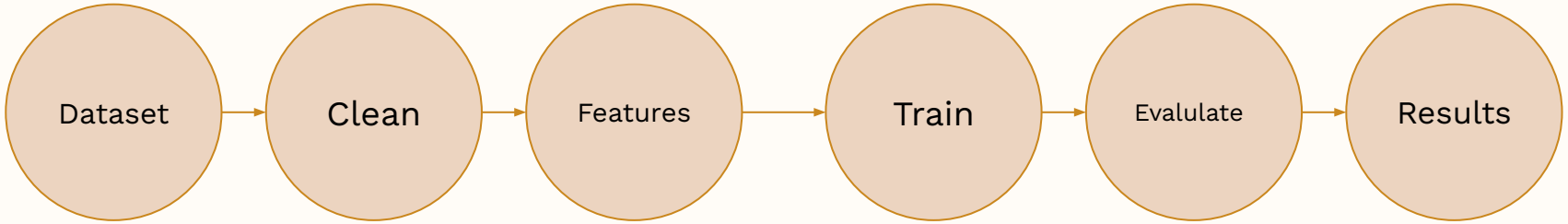
3 day rolling avg as 1 headline on single day may cause noise.

Scaling

Standardization only applied to some models that depend on feature magnitude

- Logistic Regression
- SVM

Pipeline



Models/Method used

Baseline Models and additional models

- Logistic Regression
  - Linear, simple and most interpretive
  - Benchmark for more complex models
  - StandardScaler
- Gradient Boosting
  - nonlinear relationships: technical and sentiment features
  - Resilient to noisy financial data.
- Random Forest
  - Nonlinear pattern detection
  - New tree correct old prediction errors
- SVM
  - margin-based classification
  - Easy class separation
  - StandardScaler

Evaluation

Train/test split

70/30 chronological split

- 70% Training
- 30% Testing

Avoids bias

- future data does not influence the past

Applied to all data sets

Evaluation metrics

Accuracy

- # Days Model predict correctly overall

Precision

- how often does the model correctly detect stock price increase.

Recall

- Of all the days of price increase how many did the model detect

f1

- Mean of precision and recall

roc_auc

- Model probability prediction, 0.5 = random, 1.0 = perfect

Result

	Model	Acc Base	Acc Sent	Acc Δ	Precision Base	Precision Sent	Precision Δ	F1 Base	F1 Sent	F1 Δ	AUC Base	AUC Sent	AUC Δ
0	SVM	0.5468	0.5099	-0.0369	0.5454	0.5355	-0.0098	0.6916	0.5942	-0.0975	0.4825	0.5043	0.0218
1	Random Forest	0.5509	0.5435	-0.0074	0.5523	0.5446	-0.0078	0.6758	0.6852	0.0094	0.5330	0.5193	-0.0137
2	Logistic Regression	0.5575	0.5345	-0.0230	0.5504	0.5373	-0.0131	0.7014	0.6910	-0.0104	0.5229	0.5058	-0.0171
3	Gradient Boosting	0.5460	0.5337	-0.0123	0.5503	0.5390	-0.0113	0.6687	0.6791	0.0104	0.5284	0.5071	-0.0213

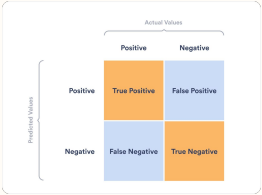
VADER

	Model	Acc Base	Acc Sent	Acc Δ	F1 Base	F1 Sent	F1 Δ	AUC Base	AUC Sent	AUC Δ
0	Random Forest	0.5509	0.5542	0.0033	0.6862	0.6978	0.0117	0.5334	0.5295	-0.0039
1	SVM	0.5460	0.5402	-0.0057	0.6940	0.6950	0.0010	0.4754	0.4660	-0.0095
2	Gradient Boosting	0.5452	0.5337	-0.0115	0.6572	0.6354	-0.0217	0.5321	0.5203	-0.0119
3	Logistic Regression	0.5591	0.5337	-0.0255	0.7015	0.6900	-0.0115	0.5228	0.5035	-0.0193

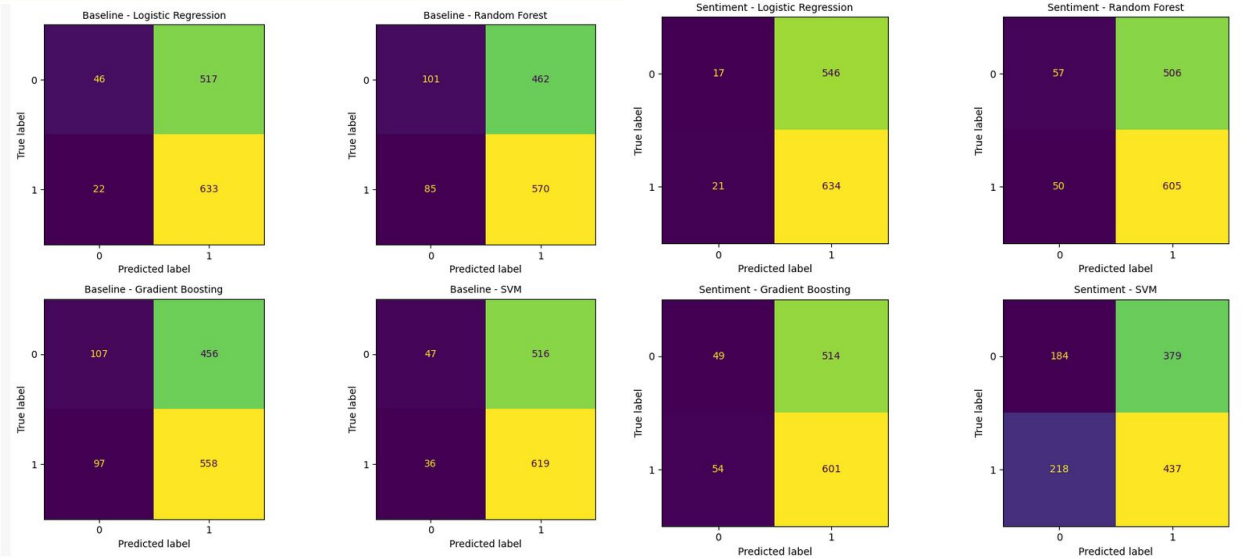
FinBERT

Comparison

Peformed the same indicating that sentiment did not improve predictions based on scoring methods used with features selected from both Vader and FinBert.

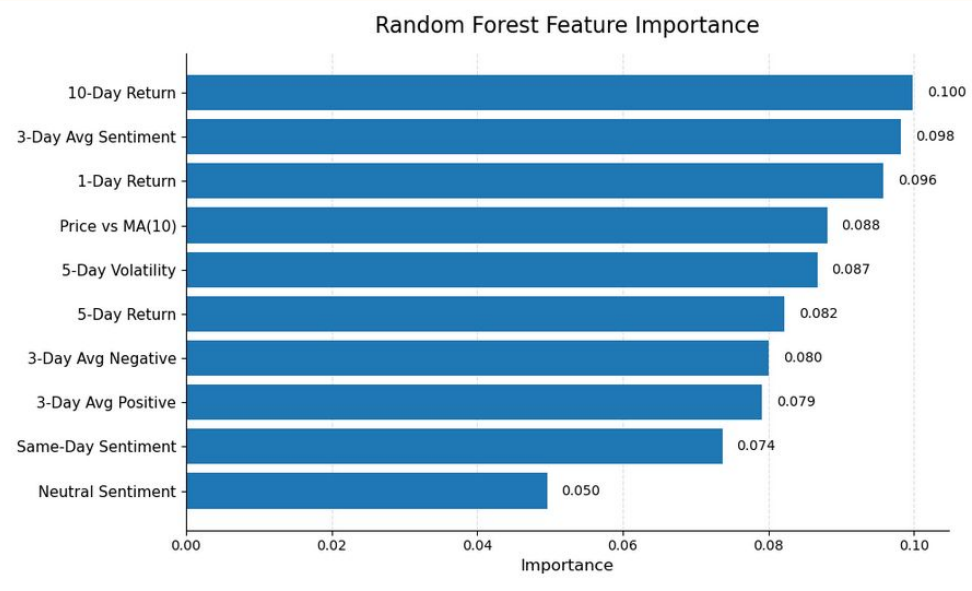


Confusion Matrix



Overall sentiment led to decreased correct amount of cases and all models favored guessing more positive.

Feature Importance



Many of the sentiment features were not as helpful in prediction.

Interpretation, Conclusion, and Future Work

Limitation and problem:

- Our sentiment scoring method may not have been nearly robust enough. Rely on 3rd party group
- Other sentiments and not just headlines
- Market movement is too noisy
- S&P 500 may be less impacted by sentiment compared to individual companies.

Best Approach:

- Random Forest, best result after adding sentiment score.

Future work and Improvements:

- Include other sentiments
- Focusing on one company